

IoT-Based Anomaly Detection for Livestock Health Monitoring Using Machine Learning

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1. Introduction

Livestock farms face major difficulties in monitoring animal health manually, especially in large farms with many animals. Farmers usually depend on visual observation to identify unhealthy animals. However,

some diseases begin with small behavioral changes that are difficult to notice early. Late detection may increase disease spread, treatment cost, and economic loss.

Recent developments in the Internet of Things (IoT) and Machine Learning provide new methods for smart livestock monitoring. Wearable collar sensors can collect continuous data such as movement activity, heart rate, and location. Machine learning models can analyze these data streams and detect abnormal behavior patterns.

This research proposes a smart livestock monitoring system that combines IoT wearable sensors with anomaly detection techniques to identify abnormal animal behavior at an early stage. The proposed system focuses on detecting unusual health-related changes rather than providing direct medical diagnosis.

The study uses the MmCows dataset, a public multimodal livestock monitoring dataset available on GitHub and introduced in NeurIPS 2024.

2. Problem Statement

Traditional livestock monitoring depends mainly on manual inspection by farm workers. This approach becomes inefficient in large farms because continuous observation of every animal is difficult and time-consuming.

In many cases, disease symptoms are discovered only after the condition becomes serious or spreads to other animals. Existing monitoring systems often focus on average herd behavior instead of individual animal behavior.

The main problem addressed in this research is the lack of an automated system capable of detecting abnormal health-related

behavior for each animal separately using real-time IoT sensor data.

3. Research Gap

Many previous studies focused on livestock tracking, activity monitoring, or supervised disease classification models that require labeled medical data. However, labeled disease datasets are limited and expensive to collect.

Few studies focus on unsupervised anomaly detection using individual behavioral baselines for each animal. This research attempts to address this gap by applying anomaly detection techniques on multimodal IoT livestock data without requiring disease-labeled training records.

4. Research Questions

- Can machine learning-based anomaly detection identify abnormal livestock behavior using IoT sensor data?
- Which sensor features contribute most to abnormal behavior detection?

5. Research Objectives

- Build behavioral profiles for individual animals.
- Detect abnormal movement and heart-rate patterns using machine learning.
- Evaluate the performance of the proposed system using standard metrics.

- Identify the most effective sensor features for anomaly detection.

6. Significance of the Study

Practical Significance

Early detection of abnormal animal behavior may reduce disease spread, lower veterinary costs, and improve livestock management efficiency.

Academic Significance

This research contributes to the application of IoT and machine learning in livestock health monitoring using real-world multimodal sensor data.

7. Scope and Limitations

Within Scope	Outside Scope
Movement and heart-rate monitoring	Medical diagnosis
IoT wearable collar sensors	Video and image analysis
Machine learning anomaly detection	Real farm hardware deployment
MmCows dataset	Other animal species

8. Dataset

Dataset Name: MmCows Dataset

The dataset contains multimodal livestock monitoring data collected from dairy cows using wearable IoT sensors.

Source: NeurIPS 2024

Available at: <https://github.com/neis-lab/mmcows>

Property	Details
Animals	10 dairy cows
Duration	14 days
Sensor Data	Movement, location, body temperature
Type	Time-series IoT data

9. Methodology

This research follows a quantitative experimental methodology. The methodology is selected because the research depends on measurable IoT sensor data and machine learning evaluation metrics.

Step 1: Data Preparation

Sensor data related to movement, location, and heart-rate behavior will be extracted and normalized for each animal separately.

Step 2: Behavioral Baseline Construction

The system will create a normal behavioral profile for every animal using historical sensor records.

Step 3: Anomaly Detection

Isolation Forest will be used as the anomaly detection algorithm because it is efficient for unsupervised anomaly detection and does not require disease-labeled training data.

Step 4: Evaluation

The system performance will be evaluated using Accuracy, Precision, Recall, and F1-Score.

10. Tools and Technologies

Tool	Purpose
Python	Machine learning implementation
scikit-learn	Isolation Forest implementation
Pandas	Data processing
NumPy	Numerical operations

11. Expected Outcomes

- Development of a livestock anomaly detection model using IoT data.
- Early detection of abnormal livestock behavior patterns.
- Identification of important behavioral features.
- Performance evaluation using machine learning metrics.

12. Timeline

Task	Duration
Literature Review	2 Weeks

Dataset Preparation	2 Weeks
Model Development	2 Weeks
Evaluation	1 Week
Writing and Revision	1 Week

13. References

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